

Coxeter Group

Note Taker: **Edward Chang**

Email: *edward.chang@uwaterloo.ca*

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1 Introduction

Def: Group

A **group** $(G, \bullet)$ where G is a set and $\bullet : G \times G \rightarrow G$ is a binary operator on the set, and it follows that:

- i) (Associativity): $\forall a, b, c \in G, (a \bullet b) \bullet c = a \bullet (b \bullet c)$
- ii) (Identity element): $\exists 1 \in G : \forall a \in G a \bullet 1 = 1 \bullet a = a$
- iii) (Inverse element): $\forall a \in G \exists a^{-1} \in G : a \bullet a^{-1} = a^{-1} \bullet a = 1$

Def:**Note**

Root vec